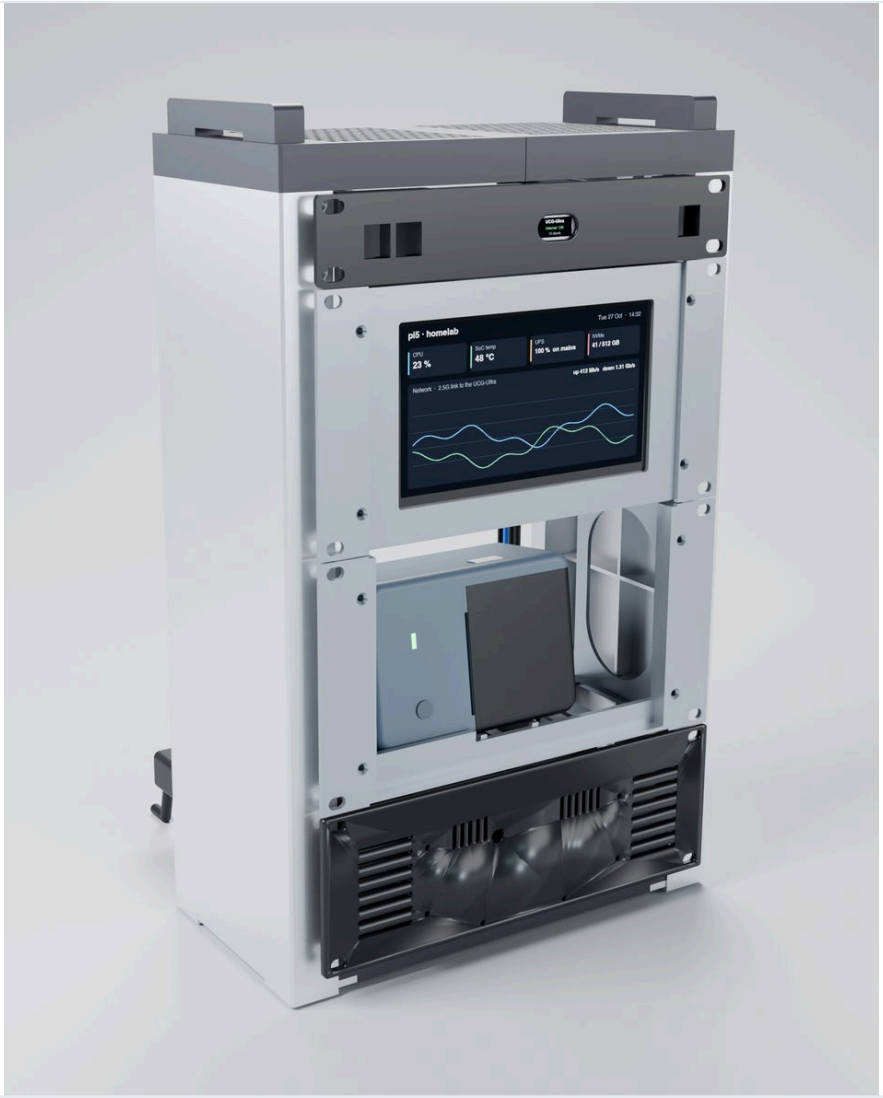
OFFICIAL INSTRUCTION MANUAL

SLIM CARGO 9U
MINI HOMELAB RACK

A precision-engineered 140 mm depth 10-inch standard rack. Designed for airline carry-on suitcases, field deployments, and portable server infrastructure with full shock-resistant interlocking modules.

DEPTH PROFILE **140 mm (-30% Volume)**	STANDARD SPEC **10-Inch EIA-310 (9U)**
TRAVEL WEIGHT **~1.24 kg (PETG Frame)**	POWER ARCHITECTURE **External 12V 5A Brick**

CRITICAL BUILD NOTICE: The 187 mm long 12V brick cradle does not fit inside this 140 mm chassis. The 12V brick operates externally, keeping the rack light and carry-on compliant.



1:1 FASTENER SIZING GAUGE (COMPARE HARDWARE HERE)

M5 × 12 mm Button Head (Allen)	48 pcs (Need)
M5 Flat Washer (Plain MS)	16 pcs (Unit ears)
M3 × 8 mm Button Head	4 pcs (Screen clips)
M3 × 8 mm Countersunk	14 pcs (Ears & Faceplate)
M3 × 12 mm Countersunk	3 pcs (Poly+ 5 Tray)
M5 Heat-Set Insert (Ø6.4 × 8–10 mm)	48 pcs (Frame & Rails)
M3 Heat-Set Insert (Ø4.0 × 5–6 mm)	18 pcs (Bodies & Shelf)

PLASTIC SPECIFICATION: Print frame in **PETG Black** (4 perimeters, 25% gyroid infill) for maximum drop resistance. Screen body in PLA Silver.

PROCUREMENT & CART VERIFICATION STATUS

Hardware	Required	In Cart / Source	Status
M5 × 12 mm Screws	48	3 pkts (75 pcs)	✅ **VERIFIED (+27)**
M5 Washers	16	1 pkt (100 pcs)	✅ **VERIFIED (+84)**
M3 × 8 mm Button	4	1 pkt (25 pcs)	✅ **VERIFIED (+21)**
M3 Heat-Set Inserts	18	1 pkt (500 pcs)	✅ **VERIFIED (+482)**
M5 Heat-Set Inserts	48	90 pc mixed pack	⚠️ **VERIFY COUNT**
M3 × 8 mm Csk	14	Amazon / Local	❌ **BUY EXTERNALLY**
M3 × 12 mm Csk	3	Amazon / Local	❌ **BUY EXTERNALLY**
M2.5 × 25 mm Pan	4	Amazon / Local	❌ **(If F-Holder passes)**

PACK WARNING: The Robu M5-M6 90-pack mixes both sizes. Count the M5 inserts upon arrival. If fewer than 48 are present, grab a standalone pack of M5 inserts before melting.

BAG 1 HEAT-SET BRASS INSERTS MASTER MAP

48 × M5 INSERTS | 18 × M3 INSERTS | SOLDERING IRON @ 230°C

TOOLS & PARTS: 48x M5 Heat-Set Inserts | 18x M3 Heat-Set Inserts | Soldering Iron + Heat Tips | Flat Surface / Tweezers

1 2U & 3U Frame Joints

Press **3 × M5 inserts** into the top mating bosses of each side module:

- 2U Base: 3 per side = 6
- Lower 3U: 3 per side = 6
- Upper 3U: 3 per side = 6

Total: 18 × M5 Inserts

2 1U Top, Cap & Handles

Install M5 inserts into the upper rack structural points:

- 1U Top ribs: 2 per side (front & rear; leave center empty) = 4
- Cap Halves: 2 per handle boss = 4

Total: 8 × M5 Inserts

3 Base Tie Rods

Take the **3 × SlimCargo_BaseBar** tie rods:

- Press **1 × M5 insert** into each end hole
- 2 inserts per bar × 3 bars

Total: 6 × M5 Inserts

4 Sub-Module Bodies

Install the smaller M3 brass inserts into printed modules:

- B1 Dock body: 4 for ears
- Screen body: 4 ears + 4 clips = 8
- Pi Shelf: 6 in front posts

Total: 18 × M3 Inserts

🎯 STEP 5: FRONT EQUIPMENT RAILS — 16 M5 INSERTS MAP

LEFT & RIGHT FRONT POSTS ONLY · REAR POSTS STAY EMPTY!

UNIT 1: Pi 5 & UPS (2U)
HOLES 1 & 6
Counted from bottom of 2U Base. 4 inserts total.

UNIT 2: B1 DOCK (3U)
HOLES 7 & 15
Directly above Pi faceplate. 4 inserts total.

UNIT 3: 7" DISPLAY (3U)
HOLES 16 & 24
Middle equipment window. 4 inserts total.

UNIT 4: UCG-ULTRA (1U)
HOLES 25 & 27
Top 1U rack rail positions. 4 inserts total.

BAG 2 CHASSIS & FRAME CONSTRUCTION

15 PRINTED PIECES | 32 × M5 × 12 MM SCREWS | INTERLOCKING PUZZLE JOINTS

PARTS FOR FRAME: 2x 2U Base | 4x 3U Modules | 2x 1U Top | 3x Base Bars | 2x Cap Halves | 2x Handles | 32x M5×12mm Screws

1 Step 2.1: The Foundation

Stand the Left and Right **2U Base Modules** upright. Bridge the bottom channels using the **3 × Base Bars** (Front, Middle, Rear). Fasten each bar from underneath with 2 × M5 × 12 mm screws (**6 screws total**).

2 Step 2.2: Stacking the Towers

Drop the side modules down sequentially onto the interlocking puzzle keys:

- Drop Lower 3U pair onto 2U Base → Fasten with 6 × M5×12mm
- Drop Upper 3U pair onto Lower 3U → Fasten with 6 × M5×12mm
- Drop 1U Top pair onto Upper 3U → Fasten with 6 × M5×12mm

Tighten all joint screws firmly to lock anti-racking trusses.

3 Step 2.3: Cap & Luggage Handles

Slide the two **Cap Halves** together via center dovetail joint. Lower onto 1U top modules and secure with **4 × M5×12mm screws**.

Place the **Low-Profile Travel Handles** into top cap recesses. Secure with 2 × M5×12mm screws each (**4 screws total**).



CHASSIS RIGIDITY CHECK: Ensure the frame sits flat without rocking. The front and middle base bars provide the floor rest for the 2U Pi shelf.

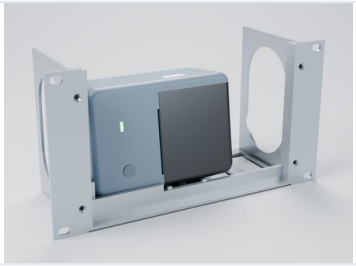
BAG 3 HOMELAB MODULE SUB-ASSEMBLIES

4 COMPLETE UNITS | 14 × M3 CSK SCREWS | 4 × M3 BUTTON SCREWS

A NIIMBOT B1 Dock (3U)

4x M3×8 Csk

1. Attach 2 × 3U Unit Ears to dock sides with 4 × M3×8 mm countersunk screws.
2. Route 50 cm USB-A to USB-C cable through rear port.
3. Seat NIIMBOT B1 printer in cradle.



Fits B1 lid swing up to 120°.

B Waveshare 7" Touchscreen (3U)

4x Csk + 4x Btn

1. Attach 2 × 3U Unit Ears with 4 × M3×8 mm Csk screws.
2. Line bezel ledge with 1 mm foam tape.
3. Seat 7" screen into body.
4. Lock corners with 4 × clips and 4 × M3×8 Button screws.



Tabs span 157 mm across.

C Pi 5 & Cuzor UPS Unit (2U)

6x M3×8 Csk + 3x M3×12

1. Stick Cuzor UPS into tray with foam tape; pin tray to shelf.
2. Fasten Poly+ 5 tray with 3 × M3×12 Csk screws.
3. Mount Pi 5 in Poly+ 5 case.
4. Plug right-angle DC extension into UPS DC-IN and thread cable through shelf's right gusset arch.
5. Secure clean faceplate with 6 × M3×8 Csk screws.



Clean front: wiring stays 100% inside.

D UniFi Cloud Gateway (1U)

Printables-942972

1. Slide UCG-Ultra gateway into 1U mount tray.
2. Snap 3 × RJ45 keystone couplers into front ports.
3. Connect rear USB-C power and patch cables before screwing into the rack frame.



Display aligns with front bezel cutout.

BAG 4 FINAL INTEGRATION & CONCEALED CABLING RISER

16 × M5×12 MM SCREWS | 16 × M5 WASHERS | CONCEALED RISER ROUTE

PARTS NEEDED: 16x M5×12mm Screws | 16x M5 Washers | 1 m Flat HDMI | 1 m Micro-USB | 1 m Slim Cat6 | Velcro Zip-Ties

1 Step 4.1: Bottom-Up Racking Order

Install modules from the bottom up to ensure structural balance and easy cable routing. Secure each module with **4 × M5 × 12 mm screws + washers**:

- **Holes 1 & 6:** Slide in 2U Pi 5 Unit (rests on front and mid base bars).
- **Holes 7 & 15:** Slide in 3U NIIMBOT B1 Dock.
- **Holes 16 & 24:** Slide in 3U 7-inch Touchscreen.
- **Holes 25 & 27:** Slide in 1U UCG-Ultra Gateway.



2 Step 4.2: Concealed Wiring Riser

Route all data and display cables up the **interior right column** (seen from the front, x = -110 mm):

- **Flat HDMI (1 m):** Adapter F → Screen HDMI. Folds behind screen.
- **Micro-USB (1 m):** Pi USB 2.0 → Screen touch/power port.
- **Slim Cat6 (1 m):** Adapter F 2.5G port → UCG-Ultra LAN 1.

Fasten bundle against inside ribs with zip-ties: unit ears conceal the cables 100%!

ZERO FRONT CABLING: When viewed directly from the front, no wires or plugs are visible. The clean faceplate keeps all power leads hidden inside.

External 12V Power Connection

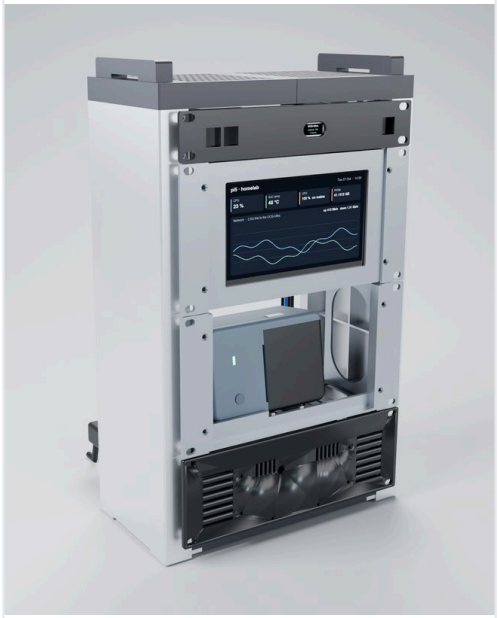
Because the 187 mm brick cradle is omitted to maintain the 140 mm slim carry-on profile, power is connected as follows:

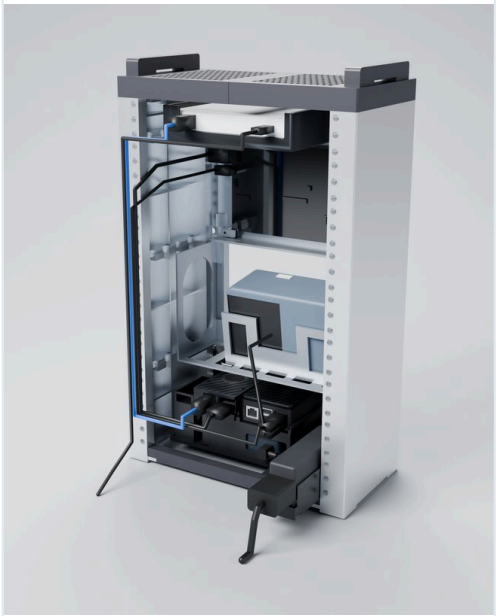
1. Desktop Setup: Place the standard 12V 5A power brick outside the rack.
2. DC Connection: Insert the brick's 5.5 × 2.1 mm barrel plug into the internal right-angle extension lead socket located inside the left bay.
3. UPS Delivery: The extension lead passes through the Pi shelf's right gusset arch directly into the Cuzor UPS DC-IN.
4. Pack & Go: When packing into a suitcase, simply unplug the external brick and stow it in your luggage accessory pouch!

Pre-Flight Software & Power Checklist

- **Pi OS USB Boost:** Add `usb_max_current_enable=1` to `/boot/firmware/config.txt` to unlock full 1.6A USB budget for the screen and B1 charging.
- **Polarity Check:** Verify center-positive (+12V on center pin) with a multimeter before initial power switch on.
- **Gateway Power:** Plug UCG-Ultra into its bundled 5V 3A USB-C adapter (the gateway runs on mains, Pi side runs on UPS).
- **Rubber Feet:** Affix 4 × stick-on rubber feet under bottom corners of the 2U base to prevent sliding on smooth surfaces.

CONGRATULATIONS! Your portable Slim Cargo 9U Homelab is fully assembled, flight-ready, and prepared for high-performance mobile edge compute!





Full Front Elevation & Rear Cable Harness Inspection Views